Jay Nasriwala

Email: jaynasriwalaoffice@gmail.com
Portfolio: https://jay-nasriwala-portfolio.vercel.app/
Mobile: +91-9313466185
Linkedin: https://www.linkedin.com/in/jay-nasriwala-674607278/
Github: https://github.com/jaynasriwala

Education

- **Sarvajanik College of Engineering and Technology** — Surat, India
  Bachelor of Technology - Computer Engineering; CGPA: 9.38 — *2022 - 2026*

Work Experience

- **Trainee Software Developer**: STPLGlobal — Surat, India
  Research & Development Department — *Jan 2026 – Present*
  - Engineered a **photogrammetry-based 3D reconstruction pipeline**, integrating **U2-Net segmentation** to achieve **~95% masking accuracy** for automated extraction from **100+ multi-view images** per scan.
  - Created a **Structure-from-Motion (SfM)** pipeline with **OpenCV** for feature extraction, matching and sparse reconstruction generating **10,000+ points**, scaling to **1,00,000+** via dense reconstruction with **OpenMVG**.
  - Optimized **initial pose estimation** by enhancing **RANSAC based image pair selection** using **SQLite driven querying** to identify **top-10 image pairs with maximum inliers**, improving reconstruction stability and achieving **~96% higher accuracy** in model initialization.
  - Calibrated **IDS industrial camera** by tuning imaging parameters: **gains, exposure, and aperture**, and performed **geometric calibration** using a checkerboard pattern to compute the **camera intrinsic matrix**.
- **Software Developer Intern**: STPLGlobal — Surat, India
  Research & Development Department — *Jan 2026 - May 2026*
  - Contributed to the **SlysX** project, developing a high-performance **3D geometry analysis and slicer engine** using **Python, Delphi, and DirectX 9**, with support generation, thickness and overhang analysis, auto-orientation, peel-force estimation, **Octree-based spatial partitioning**, ray casting, and parallel processing for meshes up to **1M triangles**.

Projects

- **CommentIQ - AI-Powered YouTube Comment Sentiment Analysis Platform**
  - Collaboratively developed an AI-powered **YouTube comment sentiment analysis platform** with a friend, implementing a multi-stage **NLP pipeline** using **Sentence Transformers**, cosine similarity, and **RoBERTa GoEmotions** for semantic filtering and fine-grained emotion detection, enhanced with Gen Z slang normalization and emoji-aware sentiment correction.
  - Built a production-ready full-stack application using **React, FastAPI, JWT authentication, SQLite caching, asynchronous job processing, and Docker**, enabling scalable audience emotion and engagement analysis. (Source Code)
- **MalariaDetect AI - AI-Powered Malaria Parasite Detection System**
  - Led a team of four under the mentorship of Prof. (Dr.) Mayuri Mehta to conduct a research project using a **EfficientNetV2-S**-based deep learning model with **Grad-CAM++** for malaria parasite detection in thin blood smear images; achieved **99% training** and **98% testing accuracy**. (Source Code)
- **Universal Database MCP - Model Context Protocol Database Connector**
  - Built a **role-restricted, multi-database MCP server** enabling LLMs to safely query **PostgreSQL, MySQL, SQL Server, and SQLite**, with **RBAC, JWT and self-hosted OAuth 2.1 authentication**, tenant-scoped row-level filtering, automatic PII masking, and 4 tools for schema discovery, safe query execution, query explanation, and multi-format export. (Source Code)

Technical Skills

- **Languages:** Python, C, Java
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Web Technologies:** React, Flask, FastAPI, RESTful APIs, Streamlit
- **AI & Agentic Systems:** MCP, RAG Pipelines, ML, DL, NLP, CrewAI, LangChain, LangGraph

Research Papers & Certifications

- Malaria Parasite Detection in Thin Blood Smear Images Using ConvNeXt with Explanations (View Paper)
- Segmenting Brain Tumor from MRI using an Efficient Encoder-Decoder-based DeepLabv3+ (View Paper)
- Fundamentals of Deep Learning By NVIDIA (View Certificate)
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional (View Badge)